

Lecture Notes on Quantum Field Theory II

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1 The Dirac Field

1.1 Lorentz Invariance

Readings

- Peskin & Schroeder Sec. 3.1
- Zee Chap. II

Relativity and Quantum Mechanics

In nonrelativistic quantum mechanics, the dynamics of a particle is described by the Schrödinger equation

$$i\frac{\partial}{\partial t}\psi(t, \mathbf{x}) = -\frac{\nabla^2}{2m}\psi(t, \mathbf{x}). \quad (1.1)$$

However this equation is not compatible with special relativity, because time and space appear differently: the time derivative appears linearly while spatial derivatives appear quadratically.

In special relativity space and time must be treated on equal footing. The fundamental symmetry of spacetime is the **Lorentz symmetry**. **Lorentz invariance** means that if the original fields satisfy the equations of motion, the transformed fields must also satisfy the same equations.

Lorentz Transformations

In special relativity we combine time and space into spacetime coordinates

$$x^\mu = (t, \mathbf{x}) \quad (1.2)$$

where $\mu = 0, 1, 2, 3$. The invariant spacetime interval is

$$s^2 = -t^2 + \mathbf{x}^2. \quad (1.3)$$

Lorentz transformations are linear transformations

$$x'^\mu = \Lambda^\mu_\nu x^\nu \quad (1.4)$$

that leave the spacetime interval invariant

$$\eta_{\rho\sigma} \Lambda^\rho_\mu \Lambda^\sigma_\nu = \eta_{\mu\nu}, \quad (1.5)$$

where the Minkowski metric in our convention is

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1.6)$$

For example, a boost in the x direction is

$$t' = \gamma(t - vx), \quad (1.7)$$

$$x' = \gamma(x - vt), \quad (1.8)$$

where $\gamma = 1/\sqrt{1 - v^2}$.

Relativistic Wave Equations

To construct relativistic wave equations we use Lorentz invariant operators. The derivative operator is

$$\partial_\mu = \left(\frac{\partial}{\partial t}, \nabla \right). \quad (1.9)$$

Using the Minkowski metric we define

$$\partial^\mu = \eta^{\mu\nu} \partial_\nu. \quad (1.10)$$

In our metric convention $(-, +, +, +)$ we obtain

$$\partial_\mu \partial^\mu = -\frac{\partial^2}{\partial t^2} + \nabla^2. \quad (1.11)$$

Using this operator we obtain the Klein-Gordon equation

$$(\partial_\mu \partial^\mu - m^2)\phi = 0. \quad (1.12)$$

This equation is Lorentz invariant and describes a relativistic scalar particle.

Limitations of Relativistic Quantum Mechanics

One may attempt to construct a relativistic quantum theory by writing wave equations for a single particle, such as the Klein-Gordon equation or the Dirac equation. However, this approach faces fundamental difficulties.

- First, in relativistic physics the number of particles is not conserved. Processes such as particle creation and annihilation can occur when the energy is sufficiently large. For example, an electron–positron pair can be created from a photon. A single-particle wave function $\psi(\mathbf{x}, t)$ cannot describe processes in which the number of particles changes.
- Second, the relativistic wave equations introduce negative-energy solutions. Interpreting these states within single-particle quantum mechanics leads to conceptual problems.

These difficulties indicate that relativistic quantum mechanics cannot be consistently formulated as a theory of a fixed number of particles. Instead, the correct framework is quantum field theory, in which fields are promoted to operators defined at each spacetime point and particles are interpreted as excitations of these fields.

We refer the reader to the discussion in Quantum Field Theory I for a more detailed explanation of these issues.

Transformation of Fields

The discussion above indicates that relativistic quantum theory must be formulated in terms of fields rather than single-particle wave functions. A field is a physical quantity defined at every spacetime point,

$$\phi(x), \quad x^\mu = (t, \mathbf{x}).$$

Since spacetime coordinates transform under Lorentz transformations, fields must transform in a consistent way so that the physical laws remain Lorentz invariant. In the following we discuss how different types of fields transform under Lorentz transformations.

Under a Lorentz transformation the coordinates change as

$$x \rightarrow x' = \Lambda x. \quad (1.13)$$

Fields transform according to their representation of the Lorentz group.

Scalar field (spin 0) A scalar field is defined as a field whose value is invariant under Lorentz transformations. This means that the value of the field at a given spacetime event does not depend on the choice of inertial frame. For a scalar field we require

$$\phi'(x') = \phi(x). \quad (1.14)$$

This equation states that the value of the field at the same spacetime event is the same in both coordinate systems. Since

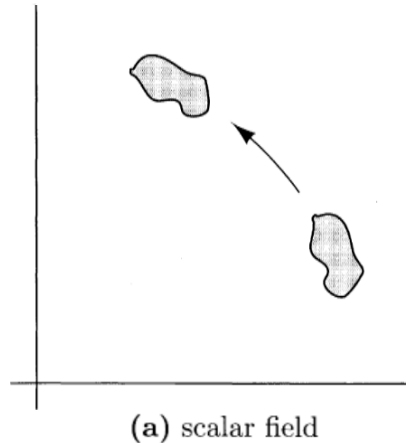
$$x = \Lambda^{-1}x',$$

we can rewrite the transformation law as

$$\phi'(x') = \phi(\Lambda^{-1}x'). \quad (1.15)$$

Finally, renaming the coordinate $x' \rightarrow x$, we obtain the commonly used form

$$\boxed{\phi(x) \rightarrow \phi'(x) = \phi(\Lambda^{-1}x)} \quad (1.16)$$



Example: Lorentz invariance of the Klein–Gordon Lagrangian Consider the Klein–Gordon Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi + \frac{1}{2}m^2\phi^2. \quad (1.17)$$

Using the scalar field transformation rule

$$\phi \rightarrow \phi'(x) = \phi(\Lambda^{-1}x), \quad (1.18)$$

we can determine how the derivatives transform. Using the chain rule we obtain

$$\partial_\mu\phi'(x) = \frac{\partial}{\partial x^\mu}\phi(\Lambda^{-1}x) = (\Lambda^{-1})^\nu{}_\mu\partial_\nu\phi(\Lambda^{-1}x). \quad (1.19)$$

Therefore

$$\partial_\mu\phi'(x)\partial^\mu\phi'(x) = (\Lambda^{-1})^\nu{}_\mu(\Lambda^{-1})^\rho{}_\sigma\eta^{\mu\sigma}\partial_\nu\phi(\Lambda^{-1}x)\partial_\rho\phi(\Lambda^{-1}x). \quad (1.20)$$

Using the property of Lorentz transformations

$$\eta_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = \eta_{\rho\sigma}, \quad (1.21)$$

one finds

$$\partial_\mu\phi'(x)\partial^\mu\phi'(x) = \partial_\mu\phi(\Lambda^{-1}x)\partial^\mu\phi(\Lambda^{-1}x). \quad (1.22)$$

Similarly,

$$\phi'^2(x) = \phi^2(\Lambda^{-1}x). \quad (1.23)$$

Therefore the Lagrangian density transforms as

$$\boxed{\mathcal{L}(x) \rightarrow \mathcal{L}'(x) = \mathcal{L}(\Lambda^{-1}x)}. \quad (1.24)$$

This means that the Lagrangian density is a Lorentz scalar. The action is

$$S = \int d^4x \mathcal{L}(x). \quad (1.25)$$

Since Lorentz transformations satisfy

$$d^4x = d^4x', \quad (1.26)$$

the action is invariant,

$$S' = S. \quad (1.27)$$

Because the equations of motion follow from the stationary condition

$$\delta S = 0, \quad (1.28)$$

the resulting Euler–Lagrange equation

$$(\partial_\mu\partial^\mu - m^2)\phi = 0 \quad (1.29)$$

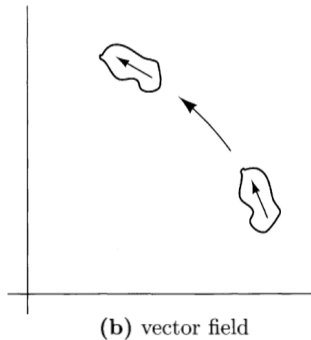
is automatically Lorentz invariant. Therefore, **if the Lagrangian density is a Lorentz scalar, the equations of motion derived from it are automatically Lorentz invariant.**

Vector field (spin 1) To understand how a vector field transforms, it is useful to first consider ordinary spatial rotations. Let $V^i(x)$ be a vector field in three-dimensional space. Under a rotation R , the coordinates transform as

$$x \rightarrow x' = Rx.$$

The vector field transforms as

$$V^i(x) \rightarrow V'^i(x) = R^i{}_j V^j(R^{-1}x). \quad (1.30)$$



This transformation has two effects.

- The argument of the field changes from x to $R^{-1}x$ because the spacetime point itself is transformed.
- The components of the vector rotate according to the rotation matrix R^i_j .

In other words, a rotation affects both the position at which the field is evaluated and the orientation of the vector itself.

The relativistic generalization of this transformation is obtained by replacing rotations with Lorentz transformations. For a four-vector field $V^\mu(x)$ we therefore have

$$\boxed{V^\mu(x) \rightarrow V'^\mu(x) = \Lambda^\mu_\nu V^\nu(\Lambda^{-1}x)}. \quad (1.31)$$

Tensor field The transformation rule for vector fields can be generalized to tensor fields. Consider a rank-two tensor field $T^{\mu\nu}(x)$. Under a Lorentz transformation, the tensor field transforms as

$$\boxed{T^{\mu\nu}(x) \rightarrow T'^{\mu\nu}(x) = \Lambda^\mu_\rho \Lambda^\nu_\sigma T^{\rho\sigma}(\Lambda^{-1}x)}. \quad (1.32)$$

Each Lorentz index introduces one factor of the Lorentz transformation matrix Λ . More generally, for a tensor with multiple indices, each index transforms independently with its own Lorentz matrix.

Index matching and Lorentz covariance Lorentz symmetry imposes a powerful constraint on the form of relativistic equations. If every term in an equation carries the same set of free Lorentz indices, then each term transforms with exactly the same Lorentz matrices. Therefore the equation preserves its form under Lorentz transformations. In this sense, matching Lorentz indices acts as a “machine” that guarantees Lorentz covariance.

E.x. Consider the equation

$$A^\mu + B^\mu = 0. \quad (1.33)$$

Under a Lorentz transformation we obtain

$$A'^\mu = \Lambda^\mu_\nu A^\nu, \quad B'^\mu = \Lambda^\mu_\nu B^\nu.$$

Both terms acquire the same Lorentz matrix, so the equation becomes

$$\Lambda^\mu_\nu A^\nu + \Lambda^\mu_\nu B^\nu = 0, \quad (1.34)$$

which is equivalent to the original equation. Therefore the equation is Lorentz covariant.

However, if the indices do not match, the equation will generally not be Lorentz covariant. For example, consider

$$A^\mu + C = 0, \quad (1.35)$$

where C is a scalar. Under a Lorentz transformation we obtain

$$A'^\mu = \Lambda^\mu_\nu A^\nu, \quad C' = C.$$

The two terms now transform differently, so the transformed equation becomes

$$\Lambda^\mu_\nu A^\nu + C = 0, \quad (1.36)$$

which is not equivalent to the original equation. Therefore the equation is not Lorentz covariant.

Spinor field (spin- $\frac{1}{2}$) Scalar, vector and tensor fields transform with powers of the Lorentz matrix Λ . However, there exist other fields whose transformation properties cannot be written in this way. These are called **spinor fields**.

A spinor field transforms as

$$\boxed{\psi_a(x) \rightarrow \psi'_a(x) = S_{ab}(\Lambda) \psi_b(\Lambda^{-1}x)} \quad (1.37)$$

The matrix $S(\Lambda)$ is a representation of the Lorentz group acting on the spinor indices. Unlike vectors, this matrix is not equal to the Lorentz matrix Λ itself. Instead it provides a different representation of the same symmetry.

Continuous symmetry groups The Lorentz symmetry belongs to a class of symmetries called **Lie groups**. A Lie group is a group of transformations that depend smoothly on continuous parameters.

Examples include

- rotations (parameter: angle θ)
- Lorentz transformations (parameters: rotation angles and boosts)
- translations (parameter: displacement a)

Representations A representation of a symmetry group is a way to realize the symmetry transformation using matrices acting on vectors.

For example, a rotation in the plane can be represented by the matrix

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (1.38)$$

Different physical objects transform according to different representations of the same symmetry group.

- scalar representation
- vector representation
- spinor representation

Generators Continuous symmetry transformations can be built from infinitesimal transformations. For example, a small rotation by an angle $d\theta$ can be written as

$$R(d\theta) = I - i d\theta J. \quad (1.39)$$

The matrix J is called the generator of the rotation.

Then, a finite transformation is obtained by composing infinitely many such infinitesimal transformations. This leads to the exponential form

$$R(\theta) = e^{-i\theta J}. \quad (1.40)$$

Thus, once the generator is known, the full continuous transformation is obtained by exponentiation.

Lie algebra The set of generators together with their commutation relations is called the **Lie algebra** of the group. The commutation relations tell us how different infinitesimal transformations combine.

For example, for spatial rotations the generators J_i satisfy the commutation relations

$$[J_i, J_j] = i\epsilon_{ijk} J_k. \quad (1.41)$$

From algebra to representation To construct a representation of the group, we first find matrices that represent the generators and satisfy the same commutation relations. Once such matrices are known, the corresponding finite transformations are obtained by exponentiation.

For example, for spin- $\frac{1}{2}$ particles we may choose

$$J_i = \frac{\sigma_i}{2}, \quad (1.42)$$

where σ_i are the Pauli matrices. Since these matrices satisfy the rotation algebra, they provide a representation of the generators. The corresponding finite rotation is then

$$U(\theta) = e^{-i\theta_i J_i} = e^{-i\theta_i \sigma_i / 2}. \quad (1.43)$$

This gives a 2×2 representation of rotations acting on spinors.

Generators of the Lorentz group The Lorentz group is also a Lie group, so its finite transformations are obtained by exponentiating its generators.

To construct the generators, it is useful to first recall the generators of spatial rotations. In three dimensions, the generators of rotations can be written as

$$J_i = i\epsilon_{ijk}x^j\partial_k. \quad (1.44)$$

These generators satisfy the rotation algebra

$$[J_i, J_j] = i\epsilon_{ij}{}^k J_k. \quad (1.45)$$

To generalize this to relativistic spacetime, we introduce antisymmetric generators

$$J^{\mu\nu} = i(x^\mu\partial^\nu - x^\nu\partial^\mu). \quad (1.46)$$

These are the generators of Lorentz transformations. Because the indices are antisymmetric,

$$J^{\mu\nu} = -J^{\nu\mu}, \quad (1.47)$$

there are six independent generators. They correspond to

- three spatial rotations
- three Lorentz boosts.

Lorentz algebra The generators satisfy the Lorentz Lie algebra

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(g^{\nu\rho}J^{\mu\sigma} - g^{\mu\rho}J^{\nu\sigma} - g^{\nu\sigma}J^{\mu\rho} + g^{\mu\sigma}J^{\nu\rho}). \quad (1.48)$$

Any matrices that represent Lorentz transformations must satisfy the same commutation relations. Therefore, any representation $M(\Lambda)$ of the Lorentz group must be constructed from matrices that obey this algebra.

Example: vector representation For a four-vector V^μ , the generators are represented by 4×4 matrices

$$(J^{\mu\nu})^\alpha{}_\beta = i\left(\delta^\mu_\beta g^{\nu\alpha} - \delta^\nu_\beta g^{\mu\alpha}\right). \quad (1.49)$$

An infinitesimal Lorentz transformation can be written as

$$V^\alpha \rightarrow \left(\delta^\alpha_\beta - \frac{i}{2}\omega_{\mu\nu}(J^{\mu\nu})^\alpha{}_\beta\right)V^\beta, \quad (1.50)$$

where $\omega_{\mu\nu}$ are small antisymmetric parameters.

Examples Different choices of the parameters $\omega_{\mu\nu}$ correspond to different Lorentz transformations.

Rotation example For a rotation in the x - y plane we take

$$\omega_{12} = -\omega_{21} = \theta. \quad (1.51)$$

The corresponding transformation acts on a vector as

$$V \rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\theta & 0 \\ 0 & \theta & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} V. \quad (1.52)$$

Boost example For a boost in the x -direction we take

$$\omega_{01} = -\omega_{10} = \beta. \quad (1.53)$$

The transformation becomes

$$V \rightarrow \begin{pmatrix} 1 & \beta & 0 & 0 \\ \beta & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} V. \quad (1.54)$$

Example: spinor representation For a Dirac spinor $\psi_a(x)$, the generators of Lorentz transformations are represented by 4×4 matrices

$$S^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu], \quad (1.55)$$

where γ^μ are the Dirac gamma matrices. These matrices satisfy the same Lorentz algebra as the generators $J^{\mu\nu}$ and therefore provide a representation of the Lorentz group acting on spinors. We will introduce more details about the spinor representations in the Dirac fields.

1.2 The Dirac Equation

Readings

- Peskin & Schroeder Sec. 3.2
- Zee Chap. II

Historical motivation

Shortly after the development of quantum mechanics by Schrödinger and Heisenberg, physicists attempted to construct a relativistic wave equation for particles. The first successful relativistic equation was the Klein–Gordon equation,

$$(\partial_\mu \partial^\mu - m^2)\phi = 0, \quad (1.56)$$

which we discussed previously. Although this equation is Lorentz invariant, it is not satisfactory as a single-particle relativistic wave equation.

Two major difficulties arise if the Klein–Gordon equation is interpreted as a wave equation.

- The probability density obtained from the Klein–Gordon current is not positive definite.
- The equation admits negative-energy solutions.

At the time these problems were viewed as technical difficulties rather than indications of a deeper issue. In 1928 Dirac attempted to construct an alternative relativistic wave equation that would overcome these problems. Dirac’s idea was to construct an equation that is

- first order in time derivatives, like the Schrödinger equation,
- Lorentz covariant,
- and reproduces the relativistic dispersion relation

$$p^\mu p_\mu = -m^2. \quad (1.57)$$

This led Dirac to propose what is now known as the **Dirac equation**, which provides a relativistic description of spin- $\frac{1}{2}$ particles such as the electron.